

Sequential Gearbox

Project Concept and Objectives

This project focuses on the complete mechanical design of a 5-speed sequential gearbox, developed as a self-driven engineering effort with the intention of approaching a realistic and manufacturable solution rather than a purely academic exercise. The gearbox architecture is based on dog-ring engagement mechanism, a configuration commonly found in high-performance and motorsport applications. A mechanism that operates through purely mechanical means.

Unlike conventional gear reducers or speed multipliers, which operate under fixed and well-defined conditions, a sequential gearbox inherently combines multiple operating cases within a single design. Each gear engagement represents a distinct load path and loading condition, affecting gears, shafts, bearings, splines, keys, and engagement components differently. A central challenge of this project was therefore to identify the most critical cases while ensuring overall coherence and consistency across the entire drivetrain.

The design was approached with a strong emphasis on classical mechanical engineering principles and established design standards. Component sizing and verification were performed using analytical methods drawn from machine design theory, bearing catalogs, and international standards, with careful attention paid to the assumptions underlying each calculation, their validity, and their impact on the final results.

Beyond obtaining numerical safety factors, the project aimed to develop a deeper understanding of how real mechanical systems are designed, including the need for iteration, trade-offs, and refinement. Multiple intermediate concepts and configurations were evaluated and ditched before obtaining the final design.

Main Objectives

- Design a complete and coherent sequential gearbox architecture
- Integrate gears, shafts, bearings, splines, keys, and dog clutches into a single functional system
- Apply classical mechanical design theory to verify critical components
- Develop a design that serves as a basis for further tests and verification, taking it from analytical to experimental validation

	16	15	14	13	12	11	
	ITEM NO.	PART NUMBER	DESCRIPTION	Material	Material Condition	QTY.	
M	1	Input_Shaft		AISI 4140	Q&T - 425 °C	1	
	2	Output_Shaft		AISI 4140	Q&T - 425 °C	1	
	3	NA6910R	Needle Bearing			2	
L	4	Gear_1_Output		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	5	Gear_1_Input		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	6	Key_Length_30		AISI 4340	Normalized	2	
	7	Reverse_Gear_Input		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	8	Reverse_Gear_Output		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	9	Dog_Ring_Rev_&_1		18CrNiMo7-6 (DIN 1.6587)		1	
	10	Idler_Gear		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	11	B27.7M - 3AM1-50	Retaining Ring			2	
	12	B27.7M - 3AM1-40	Retaining Ring			3	
	13	B27.7M - 3AM1-30	Retaining Ring			2	
K	14	SKF - NU 208 EC - 14,SI,NC,14	Cylindrical Roller Bearing			1	
	15	SKF - NU 2206 EC - 14,SI,NC,14	Cylindrical Roller Bearing			3	
	16	Key_Length_20		AISI 4340	Normalized	3	
J	17	Key_Length_28		AISI 4340	Normalized	2	
	18	Gear_2_Input		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	19	Gear_2_Output		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	20	NA5909	Needle Bearing			1	
	21	NK50_25RCT+IR45x50x25	Needle Bearing			1	
	22	Gear_3_Output		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	23	Dog_Ring_Sleeve_gear_2_3		18CrNiMo7-6 (DIN 1.6587)		1	
	24	Dog_Ring_2_&_3		18CrNiMo7-6 (DIN 1.6587)		1	
	25	Gear_3_Input		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	26	Gear_4_Input		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
H	27	NA4908R	Needle Bearing			1	
	28	Gear_4_Output		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	29	Gear_5_Input		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
G	30	Dog_Ring_4_&_5		18CrNiMo7-6 (DIN 1.6587)		1	
	31	Dog_Ring_Sleeve_gear_4_5		18CrNiMo7-6 (DIN 1.6587)		1	
	32	NK45_20R+IR40x45x20	Needle Bearing			1	
	33	Gear_5_Output		18CrNiMo7-6 (DIN 1.6587)	Case hardened	1	
	34	Spacer		42CrMo4 (DIN 1.7225)		1	

Verification and Engineering Work

The mechanical design of the sequential gearbox was verified using classical machine design methods and established engineering references. Gear geometry and strength were evaluated using AGMA-based criteria to ensure adequate resistance to bending and contact stresses under the most critical operating conditions.

Shafts were dimensioned and verified under combined torsional and bending loads. Stress concentration effects and fatigue behavior were evaluated using standard failure criteria as presented in *Shigley's Mechanical Engineering Design*, allowing identification of governing sections and critical load cases.

Bearing selection was performed using the Shigley bearing life methodology, based on duty-cycle-dependent equivalent dynamic loads. Bearings were selected from manufacturer catalogs such that their basic dynamic load ratings exceed the calculated equivalent loads.

Torque transmission elements, including keys, splines, and dog clutches, were verified against their governing failure modes using classical strength-of-materials approaches. Keys were checked in shear and crushing, splines were verified in bearing pressure following standard practice, and dog clutches were evaluated for crushing and shear of the dog teeth.

The final design is the result of multiple iterations, during which several intermediate configurations were evaluated and refined. This process highlights the iterative nature of engineering design, where minor adjustments can sometimes resolve identified issues, while in other cases a more substantial redesign is required to achieve a satisfactory solution.

What I Learned

This project provided a comprehensive learning experience in mechanical design, extending beyond isolated calculations to the development of a complete and coherent drivetrain system. One of the most significant lessons was learning how to define, justify, and apply engineering assumptions. Understanding when simplifications are valid, how they influence results, and how to interpret safety factors in context proved to be as important as the calculations themselves.

From a technical perspective, the project deepened my understanding of drivetrain components and their interaction within a complex mechanical system. Designing a sequential gearbox required integrating gears, shafts, bearings, splines, keys, and dog clutches into a single architecture, while accounting for multiple operating cases and distinct load paths. This differs fundamentally from the design of fixed-ratio reducers, where operating conditions are more limited and uniform.

In addition to technical knowledge, this work strengthened practical engineering skills related to design iteration and problem-solving. Several intermediate design concepts were analyzed and discarded as new constraints emerged, leading to progressive refinement of the gearbox layout and component sizing. This iterative process highlighted the non-linear nature of real engineering design.

From a software perspective, the project improved proficiency in CAD modeling, assembly management, and technical drawing preparation, as well as in the organization and documentation of complex engineering work. Structuring calculations, design choices, and results in a clear and traceable manner was a key outcome of the project.

Overall, this project reinforced the difference between theoretical coursework and practical mechanical design, providing a realistic view of the engineering process and significantly deepening both technical understanding and engineering judgment.